Haziq Zaman Chaudhry

Software Engineer | AI & Agentic Systems Developer

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Summary

Detail-oriented Software Engineer specializing in Full-Stack Development, Artificial Intelligence, and Machine Learning. Experienced in building end-to-end applications using React, FastAPI, Python, PyTorch, and SQL-based databases, with hands-on experience integrating Computer Vision, Explainable AI, and Retrieval-Augmented Generation (RAG) workflows. Developed and deployed multi-model AI pipelines, hybrid information-retrieval systems, and AI-assisted applications, with a strong foundation in Object-Oriented Programming, Data Structures, REST APIs, and database-driven software development. Interested in AI Engineering, Agentic AI, Backend Engineering, and intelligent software systems.

Experience

Advantura Labs 1 Month

Internee

- Developed cross-platform mobile application components using Flutter and Dart.
- Collaborated within a professional software development workflow, focusing on UI/UX implementation and state management principles.
- Gained practical exposure to version control, team collaboration, and agile development cycles.

Projects

MedScan-AI

Stack: React, Vite, FastAPI, Python, PyTorch, Torchvision, SQLite, SQLAlchemy, OpenCV, NumPy, Pandas, Scikit-learn, LangChain, Sentence Transformers, BM25, Google Gemini API, JWT Authentication, Git

- Developed a full-stack educational histopathology learning platform for breast cancer analysis, integrating React, FastAPI, PyTorch, and SQLite with deep learning models including ResNet-18 and U-Net.
- Implemented IDC detection with Grad-CAM visual explainability, generating heatmaps that highlight image regions influencing model predictions for educational interpretation.
- Designed a multi-stage nuclei segmentation and mitosis analysis pipeline using NuSeC and MiDeSeC datasets, combining B1 nuclei analysis and B2 mitosis analysis to generate structured morphological findings and visual overlays.
- Built an end-to-end RAG-based medical education chatbot using Sentence Transformers (all-MiniLM-L6-v2) embeddings, BM25 retrieval, hybrid retrieval/reranking, and Gemini-based response generation to provide grounded answers from a curated medical knowledge base.
- Developed secure JWT authentication and REST APIs with FastAPI, SQLAlchemy, and SQLite, supporting user management, image analysis, saved case records, dashboard statistics, and case-history retrieval.
- Integrated interactive analysis, case history, dashboard, chatbot, and explainability modules into a unified learning environment, with the system explicitly designed for educational use rather than clinical diagnosis.

Food-Scape | Community Recipe & Restaurant Discovery Platform

Stack: HTML, CSS, JavaScript, PHP, MySQL, XAMPP

- Developed a PHP-based web platform with user authentication, session management, dynamic dashboards, and community recipe functionality.
- Implemented secure MySQL database operations using prepared statements, handling recipe storage, retrieval, JSON-encoded ingredients and instructions, and file uploads.
- Built dynamic database-driven recipe rendering with timestamp-based sorting and interactive recipe cards.

Hotel Management System | Booking & Allocation Interface

Stack: C++, OOP, File Handling

- Developed a console-based hotel management system using C++ and Object-Oriented Programming (OOP) principles to manage four room categories with different pricing structures.
- Implemented room booking, checkout, and allocation logic with state validation to prevent occupied rooms from being reassigned.
- Built a room-status dashboard to track available and occupied rooms and implemented file handling for persistent customer records and automated billing slips.

SDLC Quest — Educational 2D Quiz Game

- Developed a 2D educational quiz game in Unity using C#, featuring multiple levels with progressive difficulty
- Designed and implemented manager scripts to handle game state, level progression, and quiz logic across scenes
- Built under a compressed development timeline (weekend turnaround) while maintaining functional, presentable gameplay for a university deadline

Skills

Languages & Frameworks

Python, React, FastAPI, PHP, JavaScript (ES6+), Dart, HTML5, CSS3, Bootstrap, Unity 2D (C#)

AI & Machine Learning:

RAG Pipelines, LangChain, Agentic Tool-Use, Prompt Engineering, Vector Embeddings, PyTorch, OpenCV, Grad-CAM, Scikit-learn

Tools & Databases

MySQL, SQLite, SQLAlchemy (ORM DB), Git, GitHub

Core SE

REST APIs, JSON, System Architecture, DBMS, Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), Manual Testing

Education

Lahore Garrison University

Bachelors inSoftware Engineering • 3.25

Software Engineering Lahore • 4 Year

https://lgu.edu.pk/